Performing Contextual and Behaviour Qualitative Analysis

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Content

- Key words
- Content (file, picture), Behaviour Profile
- Online Presence and Data Acquisition

Introduction

- Qualitative analysis in fraud prevention focuses on understanding the **context** and **behavioral patterns** of users, transactions, or entities to detect anomalies and potential fraud.

- Examines **non-numeric data** such as text, images, social media activity, and behavioral trends.

Key Concepts in Contextual and Behavioral Analysis

A. Keywords Analysis

- Involves scanning text data (emails, chats, documents) for suspicious keywords that may indicate fraud.
- Example:
  - A bank monitors customer service chats for phrases like *"bypass verification"*, *"account takeover"*, or *"urgent wire transfer"* to detect social engineering attempts.
  - In insurance fraud, keywords like *"fake injury"* or *"exaggerated claim"* in emails can trigger investigations

Key Concepts in Contextual and Behavioral Analysis

B. Content Analysis (Files, Pictures, Videos)

- Examines digital content for signs of manipulation or malicious intent.
- Techniques:
  - **Metadata Analysis:** Checks file properties (creation date, author, location) for inconsistencies.
- Example: A fraudster submits an edited invoice; metadata reveals the file was modified after the stated transaction date.
  - **Image Forensics:** Detects photo alterations (e.g., deepfakes, forged documents).
- Example: A fraudster submits a fake ID; image analysis shows pixel inconsistencies indicating tampering.

– **Textual Content Analysis:** Detects plagiarism, fake reviews, or fraudulent claims in documents.

Key Concepts in Contextual and Behavioral Analysis

C. Behavior Profiling

- Builds a baseline of normal behavior for users/systems and flags deviations.
- Examples:
  - **Banking Fraud:** A user who typically logs in from New York suddenly attempts a transaction from Russia.
  - **E-commerce Fraud:** A buyer makes multiple highvalue purchases in quick succession with different cards.
  - **Insider Threats:** An employee accesses sensitive files unrelated to their role.

Online Presence & Social Media Analysis

- Fraudsters often leave digital footprints on social media, which can be used to verify identities or detect scams. **A. Social Media Investigations**
- **Identity Verification: –** Checks if a person's claimed identity matches their online profiles.
  - Example: A loan applicant claims to be a CEO, but LinkedIn shows no such profile.
- **Fraud Detection:**
  - Scammers impersonate brands or individuals; social media analysis helps confirm legitimacy.
  - Example: A fake investment scheme promotes "guaranteed returns" on Twitter but has no official website.

Online Presence & Social Media Analysis

B. Open-Source Intelligence (OSINT)

- Gathers publicly available data (social media, forums, news) for fraud investigations.
- Example:
  - A fraudster running a Ponzi scheme is traced through forum posts bragging about "easy money."

C. Deep & Dark Web Monitoring

- Fraudsters sell stolen data on dark web markets; monitoring helps prevent breaches.
- Example: A company finds its employees’ credentials for sale on a dark web forum and resets all passwords.

Digital Forensics and Data Acquisition

- Used to investigate fraud by recovering and analyzing digital evidence.

A. Digital Forensic Process

i. **Identification:** Locate relevant data (emails, logs, hard drives). **ii. Preservation:** Secure data to prevent tampering (hash verification).

iii. **Analysis:** Examine data for fraud indicators (deleted files, hidden folders).

iv. **Reporting:** Document findings for legal proceedings.

Digital Forensics and Data Acquisition

B. Common Forensic Techniques

- **File Carving:** Recovers deleted files (e.g., a fraudster deletes transaction logs).
- **Memory Dump Analysis:** Detects malware used in financial fraud.
- **Network Forensics:** Analyzes traffic logs for unauthorized access.

Key Applications

- Detecting fraudulent transactions by analyzing user behavior.
- Identifying fake accounts or impersonation through social media analysis.
- Investigating cybercrimes using digital forensics

Real-World Fraud Detection Examples

Fraud Type	Detection Method	Outcome
Credit Card Fraud	Behavioral analysis detects unusual spending (e.g., multiple high-value purchases in different countries).	Transaction blocked.
Insurance Fraud	Image forensics proves a "car accident" photo was staged.	Claim denied.
Phishing Scam	Keyword analysis flags emails with "urgent account update" links.	Email blocked.
Identity Theft	Social media analysis shows mismatched identity details.	Account frozen

Challenges in conducting Qualitative Fraud Analysis

- **False Positives:** Legitimate transactions flagged as fraud due to unusual behavior.
- **Privacy Concerns:** Balancing fraud detection with user data protection.
- **Evolving Tactics:** Fraudsters constantly adapt, requiring updated detection models.

Questions